

Tarea Sección 6.3

3) a) $\cos \frac{\pi}{8} = \sin \frac{3\pi}{8}$

b) $\sin \frac{\pi}{4} = \cos \left(\frac{\pi}{2} - \frac{\pi}{4} \right)$

c) $\tan 1 = \cot \left(\frac{\pi}{2} - 1 \right)$

d) $\csc 0.53 = \sec \left(\frac{\pi}{2} - 0.53 \right)$

9) a) $\sin \frac{\pi}{4} - \sin \frac{\pi}{6}$

b) $\sin \frac{\pi}{12} = \sin \left(\frac{\pi}{4} - \frac{\pi}{6} \right)$

$$\frac{\sqrt{2}}{2} - \frac{1}{2} = \frac{\sqrt{2}-1}{2}$$

$$\sin \frac{\pi}{4} \cos \frac{\pi}{6} - \cos \frac{\pi}{4} \sin \frac{\pi}{6}$$

$$\frac{\sqrt{2}}{2} \left(\frac{\sqrt{3}}{2} \right) - \frac{1}{2} \left(\frac{1}{2} \right)$$

$$\frac{\sqrt{6}}{4} - \frac{1}{4}$$

15) $\cos 3 \sin(-2) - \cos 2 \sin 3 = \sin(-5)$

21) $\cot x = \sqrt{3}$, $\cos x < 0$, $\tan(x + \frac{\pi}{4})$

$$1 + \cot^2 x = \csc^2 x$$

$$1 + (\sqrt{3})^2 = \csc^2 x$$

$$\sqrt{4} = \csc x$$

$$\csc x = 2$$

$$\sin x = \frac{1}{2}$$

$$\sin^2 x + \cos^2 x = 1$$

$$\cos^2 x = \frac{3}{4}$$

$$\cos x = \frac{\sqrt{3}}{2}$$

$$\frac{\tan x + \tan \frac{\pi}{4}}{1 - \tan x \tan \frac{\pi}{4}}$$

$$\frac{\sin x + \tan \frac{\pi}{6}}{\cos x}$$

$$\frac{1 - \left(\frac{\sin x}{\cos x} \right) \tan \frac{\pi}{6}}{1 - \left(\frac{\sin x}{\cos x} \right) \tan \frac{\pi}{6}}$$

$$\frac{\frac{\sqrt{3}}{3} + \frac{\sqrt{3}}{3}}{1 - \left(\frac{\frac{\sqrt{3}}{3}}{\frac{\sqrt{3}}{3}} \right) \left(\frac{\sqrt{3}}{3} \right)}$$

$$\frac{\frac{2\sqrt{3}}{3}}{\frac{2}{3}}$$

$$= \frac{2\sqrt{3}}{3} \cdot \frac{3}{2} = \sqrt{3}$$

25) a) $\sin(\alpha + \beta)$

b) $\tan(\alpha + \beta)$

$$n = \sqrt{(5)^2 - (4)^2}$$

$$\frac{(3/4) + (-4/3)}{1 - (3/4)(-4/3)} = \frac{-7/24}{1 - (-1)} = \frac{-7/24}{2} = -7/48$$

$$(-4/5)(3/5) + (3/5)(4/5) = \frac{-24/25}{1} = -24/25$$

c) cuadrante IV

33) $\cos(\theta - \pi) = -\cos \theta$

$$\begin{aligned} \cos \theta \cos \pi + \sin \theta \sin \pi &= -\cos \theta \\ -\cos \theta &= -\cos \theta \end{aligned}$$

44) $\frac{1}{\cot \alpha - \cot \beta} = \frac{\sin \alpha \sin \beta}{\sin(\beta - \alpha)}$

$$\frac{1}{\frac{\cos \alpha}{\sin \alpha} - \frac{\cos \beta}{\sin \beta}} = \frac{\sin \alpha \sin \beta}{\sin(\beta - \alpha)}$$

$$\frac{1}{\cos \alpha \sin \beta - \cos \beta \sin \alpha} = \frac{\sin \alpha \sin \beta}{\sin(\beta - \alpha)}$$

$$\frac{\sin \alpha \sin \beta}{\cos \alpha \sin \beta - \sin \alpha \cos \beta} = \frac{\sin \alpha \sin \beta}{\sin(\beta - \alpha)}$$

$$\frac{\sin \alpha \sin \beta}{\sin(\beta - \alpha)} = \frac{\sin \alpha \sin \beta}{\sin(\beta - \alpha)}$$